Technical Log

Music Analysis Tool

Overview

Import an audio file and return key signature, time signature, and tempo.

Design Limitations

Feature	Status	Notes
Tempo	Reliable	Less accurate on songs with large tempo shifts
Key	Reliable	Standard algorithm struggles with minor keys and modes; the custom system improves this
Time Signature	Limited	Overall unreliable, especially for synthesized or melodic audio. May try to implement the future

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Known Issues

- Krumhansl-Schmuckler frequently confuses relative major/minor keys
- Time signature detection is fundamentally difficult without strong rhythmic accents
- Myxolydian and Phrygian are typically identified together, but this is one-directional (Myxolydian scale gets flagged as Phrygian, but Phrygian scales are correctly identified)
- Locrian has issues being identified using a pure scale.
- Lack of songs to test all modes; accuracy using real audio is not currently guaranteed.

Custom Key Detection

Built a custom profile-based key estimator using Pearson correlation. Returns top 3 results with confidence scores—Outperforms music21's built-in Krumhansl-Schmuckler on several test cases, particularly for harmonic minor keys. Currently implements 108 profiles across 9 scale types: Major, Natural Minor, Harmonic Minor, Dorian, Phrygian, Lydian, Mixolydian, Phrygian Dominant, and Locrian (12 keys each).

Future extensions: Chord progression generation, Chord progression identification, Time signature identification

Dev Log

Development Log

5/29/26 — Initial Setup & Core Features

Setup

- Created `test.py` to verify library imports and versions
- Resolved Python interpreter mismatch between terminal and VS Code

Features Implemented

- Tempo detection via `librosa.beat.beat_track` + interval method as backup
- Smoothed tempo curve over time using `scipy.ndimage.uniform_filter1d`
- Chromagram visualization — heatmap and bar chart
- Key detection via Krumhansl-Schmuckler (music21)
- Custom key detection system — 36 profiles across major, natural minor, and harmonic minor

5/30/26 — Fixing Errors and Extending Features

Errors

- Resolved Logic error within the base c_natural_minor that gave weight to A instead of G#
- Updated logic for custom key identities: 1st, 3rd, and 5th notes are given 6.0 weighting, other notes in the mode are given 3.0, and the rest are given 1.0

Features Implemented

- Added new sample scales to test differences between librosa detection and my custom-built key detection
- Dorian, Phrygian, Locrian, Lydian, Mixolydian, and Phrygian Dominant were implemented into custom key detection
- Adjusted Locrian profile — diminished 5th (F#), weight reduced from 6.0 to 3.0 to reflect the instability of the interval. Correctly identifies C Locrian after adjustment. However, this adjustment then identifies C Dorian as C Locrian. Reverted back to 6.0

Current Limitations

- Mixolydian/Phrygian Ambiguity – Modes with the same parent scale (e.g., C Mixolydian and A Phrygian) produce nearly identical scores when tested using a scale

- Locrian – Unreliable detection due to diminished 5th
- **General note** — Chromagram-based key detection fundamentally cannot detect tonal center, only pitch content.
- Lack of proper songs to test all modes.

Test Results

5/29/26 - Initial Tests (Major, Minor, Harmonic Only)

Song	Known Key	Known BPM	Custom BPM	Custom Key	K-S Result	Correct?
老張 - No Party for Cao Dong	E Major	135	135.31 - 135.99	E Major	G# Minor Alt: B Major	✅
Tank! - The Seatbelts	C Minor	137-140	135.99 - 137.52	C Minor Alt: C Harmonic Minor	C Major Alt: C Minor	✅

Notes:

- Custom system outperforms K-S on both tests
- Tank! - The Seatbelts
    - Key is ambiguous between sources (Spotify vs Wikipedia)
    - Lower confidence scores overall reflect the harmonic complexity of the jazz arrangement.
    - Tempos are slightly off from 140.
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- 老張 - No Party For Cao Dong
    - Custom system correctly identified E major with high confidence (0.846).
    - Krumhansl-Schmuckler returned G# minor, the relative minor of E major.